

Class 3: Matrix Equations

A is a matrix with m rows and n columns denoted by

and $\mathbf{x}$ is a column vector with n entries denoted by $\mathbf{x} =$

We define the matrix product to be $A\mathbf{x} =$

In other words the matrix product $A\mathbf{x}$ is

Key idea: A matrix times a column vector is the same as a linear combination of the columns of the matrix.

The matrix equation $A\mathbf{x} = \mathbf{b}$ has a solution exactly when it is possible to write $\mathbf{b}$ as

Question: When will it be true that the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every vector in $\mathbf{R}^n$?

Example: Let $A = \begin{bmatrix} 0 & 5 \\ 1 & 2 \\ -3 & 1 \end{bmatrix}$

Example: Let $A = \begin{bmatrix} 0 & 5 & 3 \\ 1 & 2 & 1 \\ -3 & 1 & 2 \end{bmatrix}$

Solve the equation $A\vec{x} = \vec{b}$

Theorem: The following statements are logically equivalent for any $m \times n$ matrix A .

a.

b.

c.

d.

Theorem: If A is an $m \times n$ matrix, $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbf{R}$ and c is a scalar, then

a.

b.